Association for Project Management

Version #	Date	Details of change	Page / slide no.

Resource management

Learning objective:

21. Understand resource management as the ability to identify and schedule the required internal and external resources.

21a) Know how to determine the resources required and their availability to deliver activities within a project.

Resource requirements

Establishing resource requirements for the project requires several considerations:

- life cycle approach
- activities and tasks to be completed
- types of resources needed to support activities
- skill sets and capabilities needed from resources
- an understanding of what the critical tasks for resources are
- timeframe that resources are needed for
- needs of the project versus needs of business as usual (BAU)
- any external resources needed

Resources may:

- Drive project pace by placing constraints on what can be done when.
- Constrain project logistics through space needed to operate and store things.

Photo credit: Markus Winkler

Project managers work with resource or BAU managers to secure resources at the required times for the schedule.

Resource or BAU managers may provide line management to the resources in BAU and therefore may be allocated to the project.

If the resource has come from a supplier, an external resource manager may be allocated to manage relations.

21b) Understand how an organisational breakdown structure is used to create a responsibility assignment matrix (RACI).

Creating a responsibility assignment matrix

- An organisational breakdown structure (OBS) details the organisation for the project and the resources available.
- Tasks can then be allocated to the resources defined in the OBS, helping to clarify roles for people. An OBS also makes initial escalation routes clear, which helps with any resource problems that may need escalating.
- For each task, the project manager will decide if the team member is responsible, accountable, consulted, or informed (RACI), creating a responsibility assignment matrix (RAM) for the project.
- The RAM is a communication device to ensure that the people involved in doing the work are aware of the work they have to do and their position in the project organisation.
- Resource or BAU managers (these may be line managers) also consult the RAM to understand and agree the work that has been allocated to their people.

In summary, the RAM provides an understanding of the tasks or deliverables, the specific responsibilities defined within the project procedures, and the level of accountability or contribution expected from named roles or individuals within the project.

RACI

A common coding structure applied to a RAM is called the RACI matrix:

R — **Responsible**
Does the work and owns any problems.

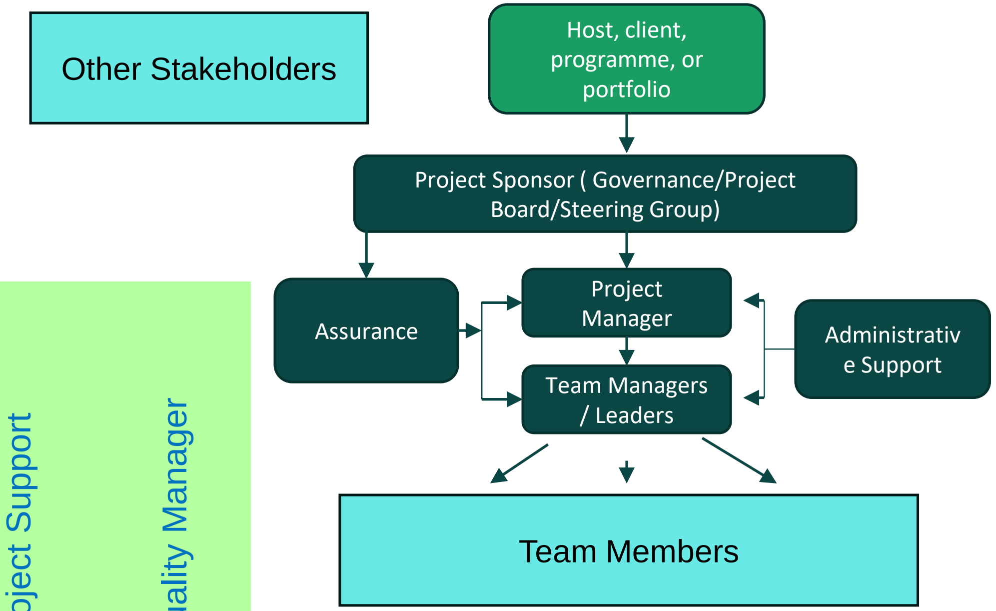
A — **Accountable**
Approves the works and is held accountable for the deliveries.

C — **Consulted**
Included in decision making and provides a key supportive role.

I — **Informed**
Kept up-to-date with progress and results.

It is best practice for one person to be responsible and one person to be accountable.

Responsibility Assignment Matrix (RAM)



From OBS

From WBS

	Sponsor	Project Manager	Team Manager	Team Member 1	Team Member 2	Team Member 3	Project Support	Quality Manager
WP A	A	R	C				C	C
WP B		A	R	C	C		I	
WP C	A	R					C	C
WP D		A	R		C	C		
WP E	I	A	R	C		C	I	I

Key:

R – Responsible
A – Accountable
C – Consult
I – Inform

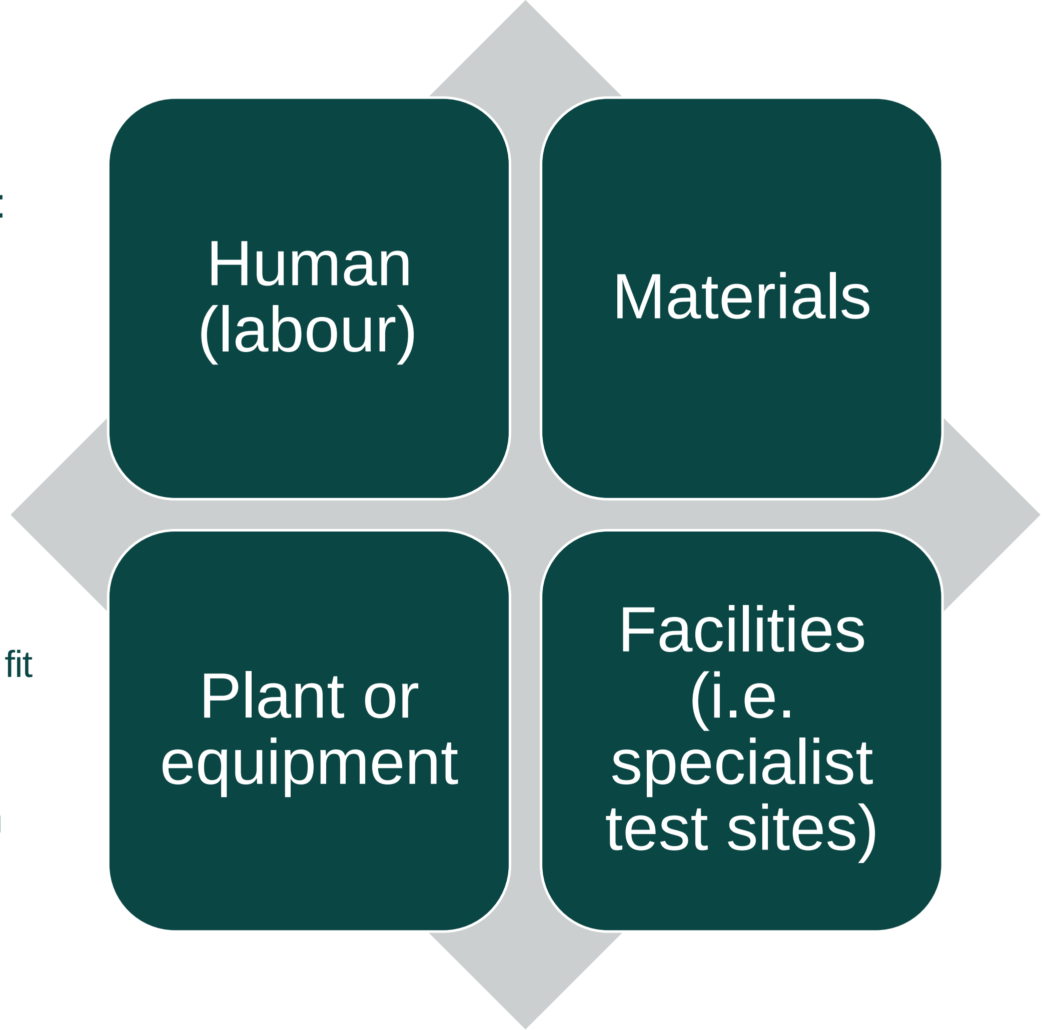
21c) Understand how resources are categorised and allocated to both linear and iterative life cycle schedules.

Categorising and allocating resources

Resources are the labour and non-labour items needed to deliver the project objectives successfully to the agreed acceptance criteria.

Resources are categorised into four types: human, materials, plant or equipment, and facilities.

- When looking to allocate resources, the project manager must consider:
  - the allocation of resource type
  - how many resources needed for each type
  - availability to support the schedule
  - cost
- When using a linear life cycle, a planning assumption made when creating the schedule is that all resources will be available when required.
  - If this is not the case, resource management techniques can be applied to try and fit the work in with the resources available as much as possible maintaining the schedule and quality.
- For iterative life cycles, requirements are prioritised and delivered within the timebox using the pre-allocated resources available, varying scope and quality as required.
  - However, if all scope must be achieved to the agreed quality, an extension of time will be needed.



21d) Know the differences between resource smoothing and resource levelling.

Resource smoothing and resource levelling

Resource management techniques, such as levelling and smoothing, attempt to fit the scope of work into the resource limit, while as much as possible maintaining the schedule and quality objectives.

- **Resource levelling** is used where a fixed amount of resource is available, and the end date of the project can be delayed. It answers the question With the resources available, when will the work be finished?

- Levelling can be achieved through redefining scope and/or specifications, increasing task durations, increasing resources earlier on to bring activities forward, or moving non-critical activities, and even critical path activities, to a different time.

- **Resource smoothing** is used when time is more important than cost, answering the question What resources do I need to deliver the work within the fixed timescale? It looks to protect the end date of the project.

- Smoothing can be achieved through reducing task durations, by adding more resources, avoiding peaks and troughs of resource demand, or changing the order of activities where the logic used originally was optional to run activities in parallel rather than in sequence.

- The overall aim of the techniques is to optimise resources through the flow of work between them, thus preventing the need to change the planned end date of the project. However, if resources are finite (either the people or equipment needed to complete the tasks) then there may be no option but to extend project timelines.

Resourced Gantt Chart & Histogram (pre-smoothing)

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Activity A																			
Activity B																			
Activity C																			
Activity D																			
Activity E																			
Activity F																			
Activity G																			
Activity H																			
Cumulative Resources	1	2	4	6	8	10	12	15	18	20	22	24	25	26	27	28	29	30	31
Total Resource/Day	1	1	2	2	2	2	2	3	3	2	2	2	1	1	1	1	1	1	1
Resource 3								F	F										
Resource 2			G	G	G	G	G	G	G	G	H	H							
Resource 1	A	A	B	B	B	B	B	C	C	C	C	C	C	D	D	E	E	E	E

Reduce the Peaks & Troughs of the profile to make it easier to manage whilst keeping to the agree timeframe

Smoothed Gantt Chart

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Activity A																			
Activity B																			
Activity C																			
Activity D																			
Activity E																			
Activity F																			
Activity G																			
Activity H																			
Cumulative Resources	1	2	4	6	8	10	12	14	16	18	20	22	24	26	27	28	29	30	31
Total Resource/Day	1	1	2	2	2	2	2	2	2	2	2	2	2	2	1	1	1	1	1
Resource 3																			
Resource 2			G	G	G	G	G	G	G	G	H	H	F	F					
Resource 1	A	A	B	B	B	B	B	C	C	C	C	C	C	D	D	E	E	E	E

Focus is on maintaining the agreed timescale

Resourced Gantt Chart & Histogram (pre-levelling)

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Activity A																			
Activity B																			
Activity C																			
Activity D																			
Activity E																			
Activity F																			
Activity G																			
Activity H																			
Cumulative Resources	1	2	4	6	8	10	12	15	18	21	24	26	26	28	29	30	31	32	33
Total Resource/Day	1	1	2	2	2	2	2	3	3	3	3	2	1	1	1	1	1	1	1
Resource 3								F	F	F	F								
Resource 2			G	G	G	G	G	G	G	G	H	H							
Resource 1	A	A	B	B	B	B	B	C	C	C	C	C	C	D	D	E	E	E	E

Current schedule requires a maximum of three resources for the 4 week period (8 to 11)

However, we get informed that our third resource has been moved and we have to manage with two people

Resource Levelling – Activity F has slipped

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Activity A																				
Activity B																				
Activity C																				
Activity D																				
Activity E																				
Activity F																				
Activity G																				
Activity H																				
Cumulative Resources	1	2	4	6	8	10	12	14	16	18	20	22	24	26	28	29	30	31	32	33
Total Resource/Day	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	1	1	1	1	1
Resource 3																				
Resource 2			G	G	G	G	G	G	G	G	H	H	F	F	F					
Resource 1	A	A	B	B	B	B	B	C	C	C	C	C	C	D	D	F	E	E	E	E

In order to overcome the Resource Constraint we have had to move activity F BEYOND the limit of the Float thus causing the overall time to extend to 20 weeks

Resource smoothing and resource levelling

Aspect	Resource Smoothing	Resource Levelling
Definition	Adjusting activities without changing the project duration	Adjusting activities based on resource constraints, which may extend the project duration
Primary Focus	Maintaining a consistent level of resource usage	Ensuring resource constraints are not violated
Schedule Impact	Low, as project end date is not changed	Medium to High, as project end date can change
Resource Utilisation	Optimises resource utilisation by minimising peaks and troughs	Balances resource demand with available supply
Critical Path	Not affected, as adjustments are within float limits	May change, as task adjustments can affect the critical path
Flexibility in Task Scheduling	Limited to activities with available float, doesn’t touch critical path activities	Flexible, can adjust any activity including those on the critical path
Usage Context	Used when resource availability is consistent and project deadlines are strict	Used when resource availability varies and meeting resource constraints is critical